

```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'NG_FFCutsFits.mat');
```

```
if ~exist('NG_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    NG_FFCutsFits = ld.fit_src_struct;
end
```

```
fn = fieldnames(NG_FFCutsFits);
nf = 11
```

```
nf =
11
```

```
%for i = 11:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei400_off200NGminFF'
```

```
selectCuts =
'Ei400_off200NGminFF'
```

```
if contains(selectCuts, 'msk')
    error('Can not fit field: %s', selectCuts);
end
if contains(selectCuts, 'all_fit_par')
    error('Can not fit field: %s', selectCuts);
end

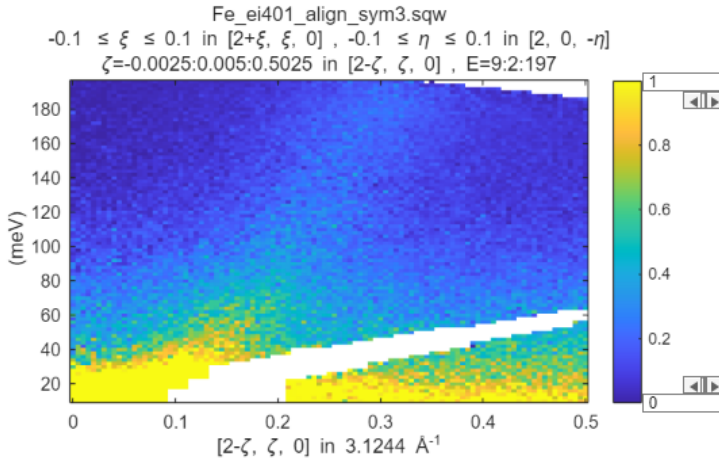
fprintf('Processing cuts: %s \n', selectCuts);
```

```
Processing cuts: Ei400_off200NGminFF
```

```
%selectCuts= 'Ei400_off200GPmaxFF';
cuts_list = NG_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts, 'Ei200')
    mi = maps_instrument(200, 600, 'S');
end
if contains(selectCuts, 'Ei400')
    mi = maps_instrument(401, 600, 'S');
end
if contains(selectCuts, 'Ei800')
    mi = maps_instrument(786, 600, 'S');
end
sample=IX_sample(true, [1,0,0], [0,1,0], 'cuboid', [0.04,0.03,0.02]);
for j=1:numel(cuts_list)
```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "NG";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

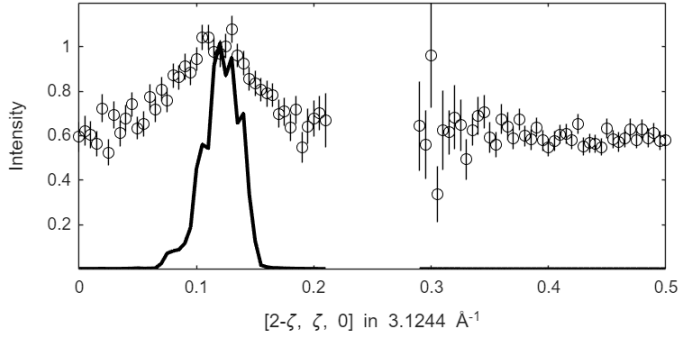
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
NG_FFCutsFits.(selectCuts) = cuts_list;

NG_FFCutsFits = fit_multicutsFF_along_direction(...
    NG_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

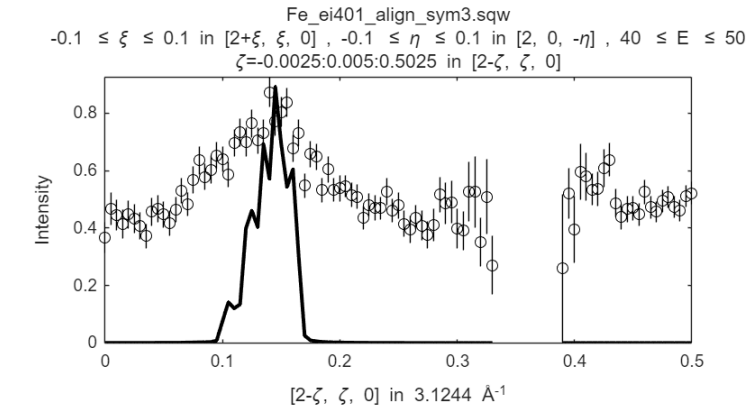
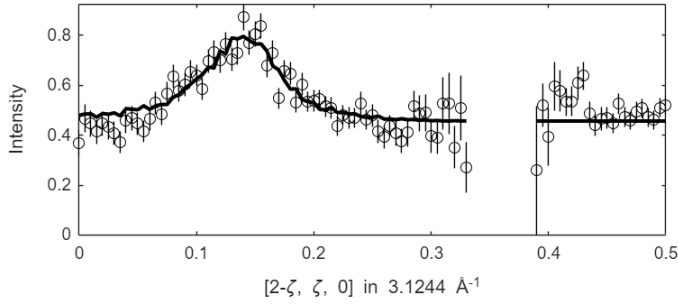
```
*****
**** En = 35±5 going to 185
*****
```

Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $30 \leq E \leq 40$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



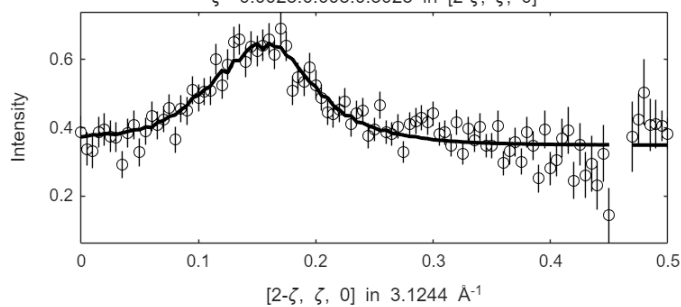
 **** En = 45±5 going to 185

Fe_ei401_align_sym3.sqw
 $S = 0.5 \pm 0.031$; $J_0 = 24 \pm 0.16$; $J_1 = 0 \pm 0$; $\gamma = 0.321723 \pm 0.0178467$
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $40 \leq E \leq 50$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

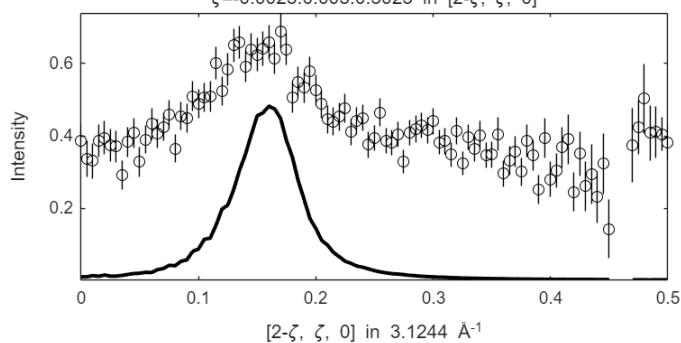


 **** En = 55±5 going to 185

Fe_ei401_align_sym3.sqw
 $S=0.78 \pm 0.1$; $J_0= 26 \pm 0.75$; $J_1= 0 \pm 0$; $\gamma=12.0829 \pm 3.628$
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $50 \leq E \leq 60$
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

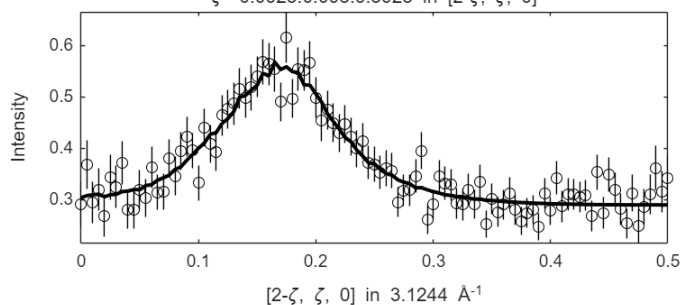


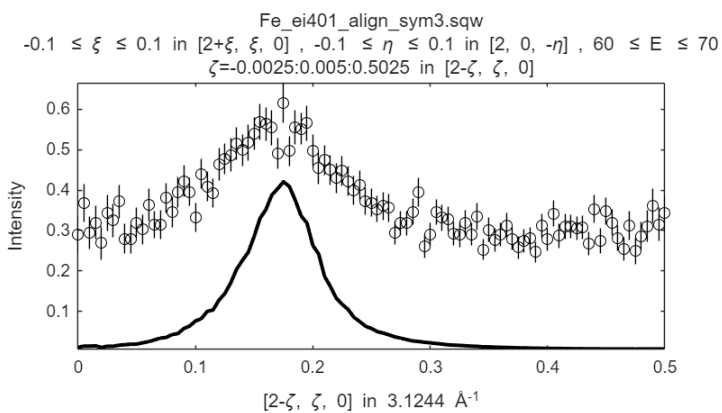
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $50 \leq E \leq 60$
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



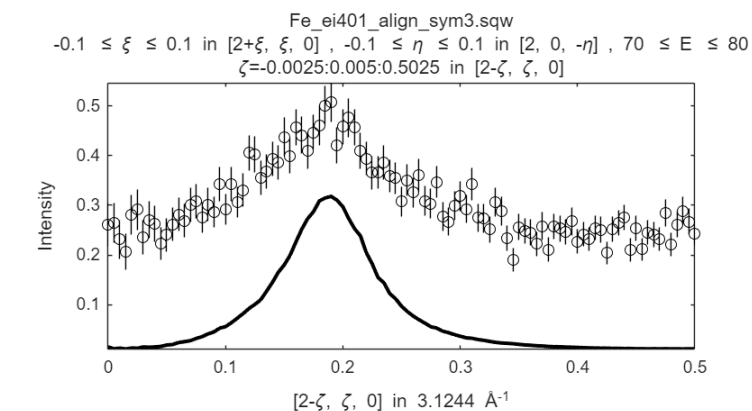
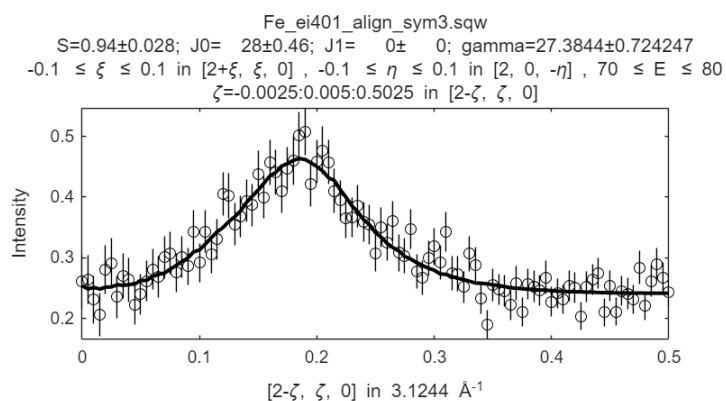
 **** En = 65 ± 5 going to 185

Fe_ei401_align_sym3.sqw
 $S=0.89 \pm 0.027$; $J_0= 27 \pm 0.47$; $J_1= 0 \pm 0$; $\gamma=18.295 \pm 0.535096$
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $60 \leq E \leq 70$
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



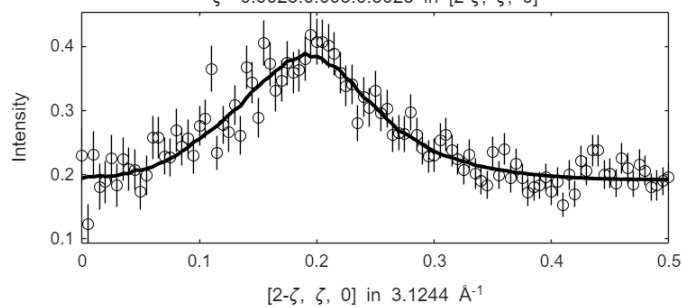


 **** En = 75 ± 5 going to 185

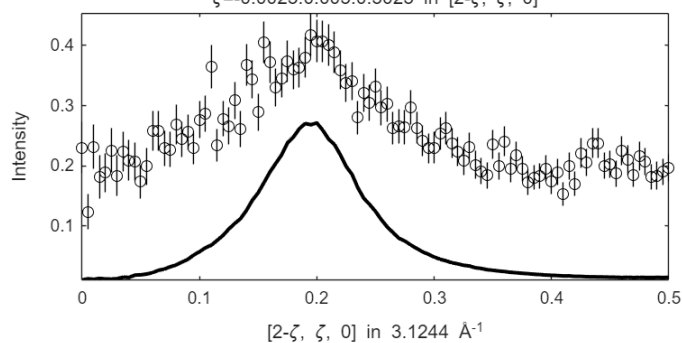


 **** En = 85 ± 5 going to 185

Fe_ei401_align_sym3.sqw
 $S = 1.1 \pm 0.19$; $J_0 = 32 \pm 1.6$; $J_1 = 0 \pm 0$; $\gamma = 38.8482 \pm 9.74608$
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $80 \leq E \leq 90$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

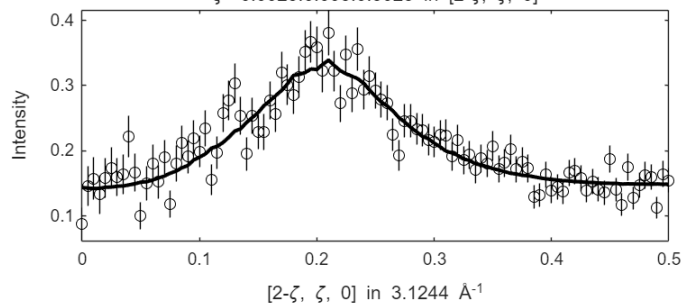


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $80 \leq E \leq 90$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

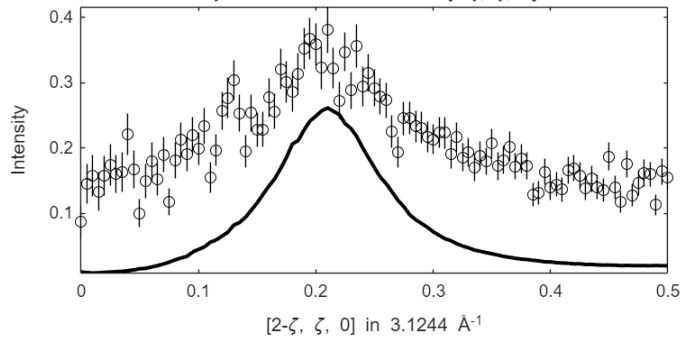


 **** $E_n = 95 \pm 5$ going to 185

Fe_ei401_align_sym3.sqw
 $S = 1.1 \pm 0.044$; $J_0 = 31 \pm 0.5$; $J_1 = 0 \pm 0$; $\gamma = 43.6681 \pm 1.24728$
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $90 \leq E \leq 100$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

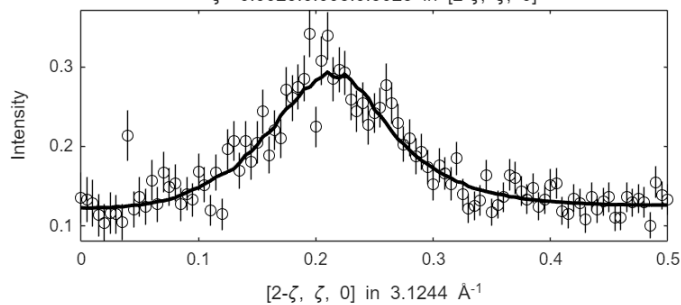


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $90 \leq E \leq 100$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

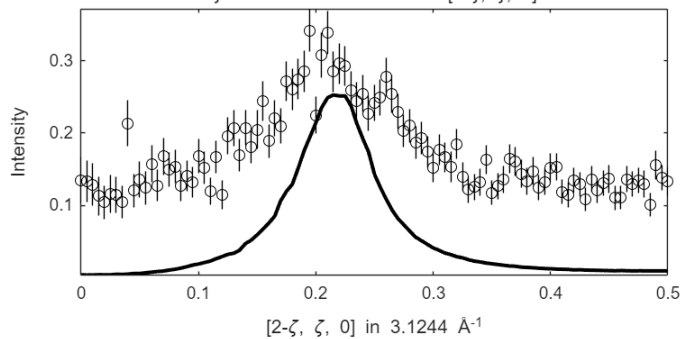


**** En = 105±5 going to 185

Fe_ei401_align_sym3.sqw
 $S=0.72 \pm 0.025$; $J_0 = 31 \pm 0.47$; $J_1 = 0 \pm 0$; $\gamma = 26.0447 \pm 0.81667$
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $100 \leq E \leq 110$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

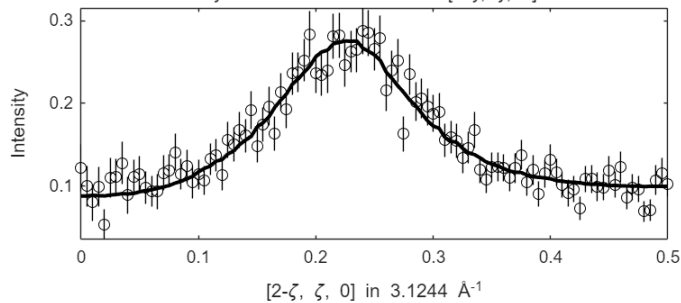


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $100 \leq E \leq 110$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

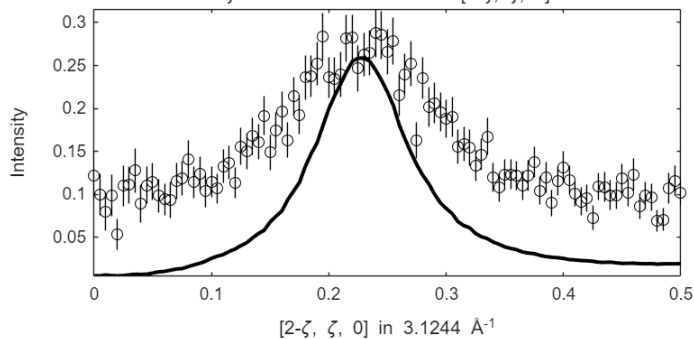


**** En = 115±5 going to 185

Fe_ei401_align_sym3.sqw
 $S=0.99\pm0.11$; $J_0= 32\pm0.81$; $J_1= 0\pm 0$; $\gamma=38.1912\pm6.26527$
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $110 \leq E \leq 120$
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

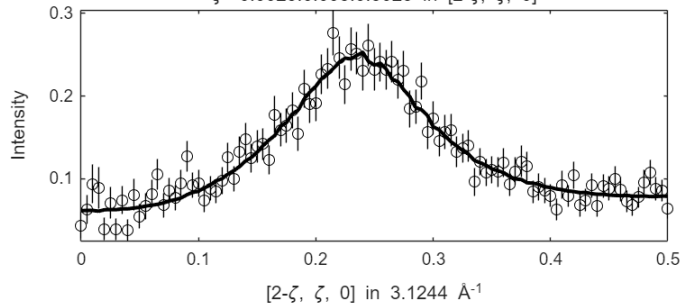


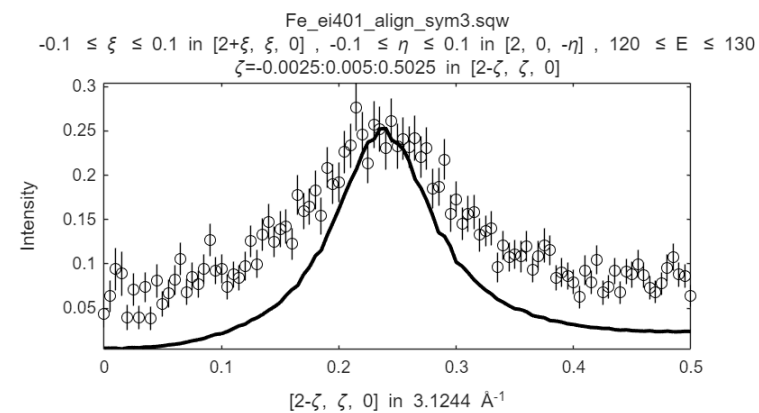
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $110 \leq E \leq 120$
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



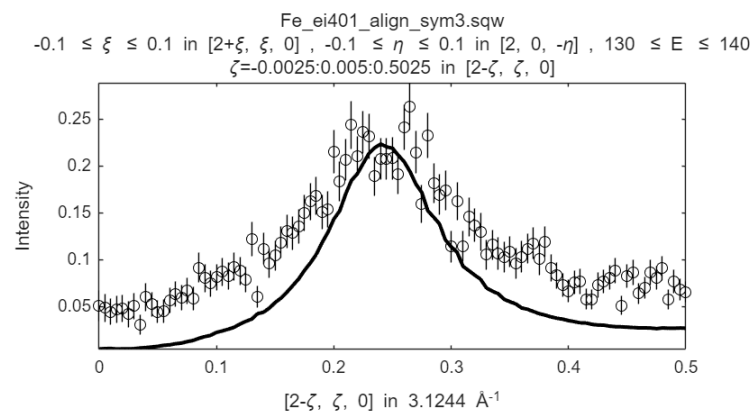
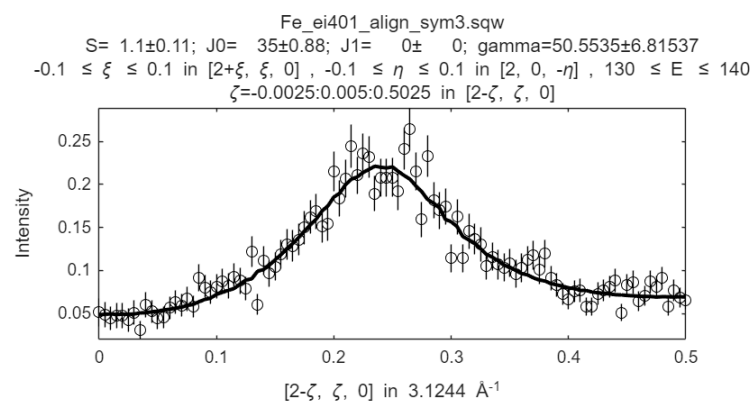
 **** En = 125±5 going to 185

Fe_ei401_align_sym3.sqw
 $S= 1\pm0.099$; $J_0= 33\pm0.72$; $J_1= 0\pm 0$; $\gamma=42.2201\pm5.67202$
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $120 \leq E \leq 130$
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



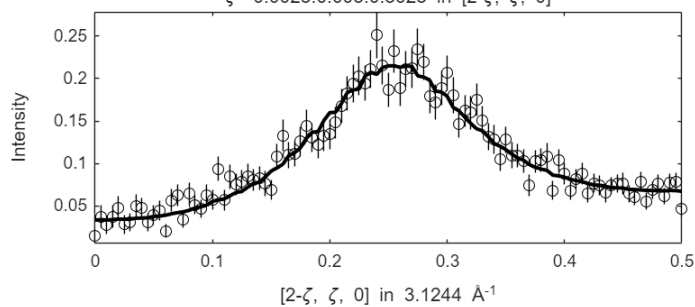


**** En = 135±5 going to 185

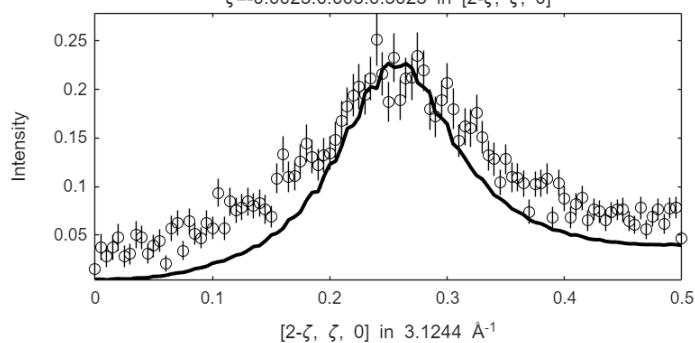


**** En = 145±5 going to 185

Fe_ei401_align_sym3.sqw
 $S = 1.2 \pm 0.066$; $J_0 = 35 \pm 0.37$; $J_1 = 0 \pm 0$; $\gamma = 56.7124 \pm 3.11743$
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $140 \leq E \leq 150$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

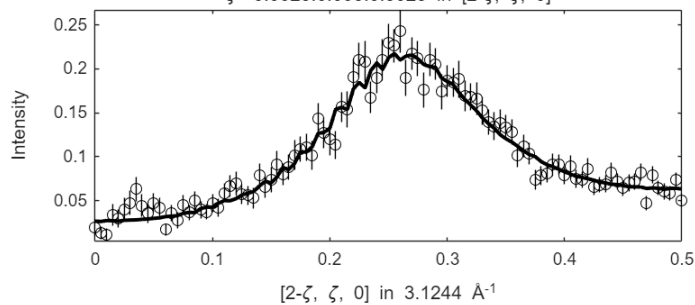


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $140 \leq E \leq 150$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

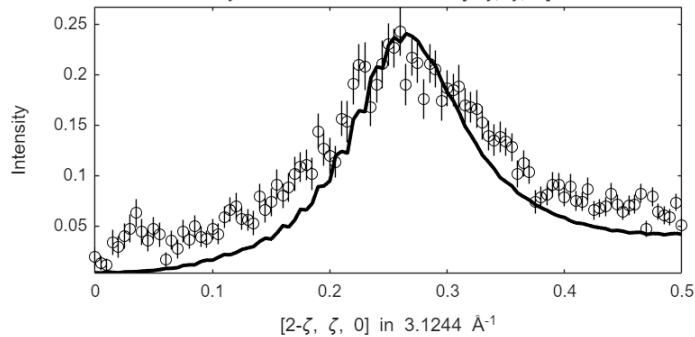


 **** En = 155±5 going to 185

Fe_ei401_align_sym3.sqw
 $S = 1.2 \pm 0.073$; $J_0 = 35 \pm 0.56$; $J_1 = 0 \pm 0$; $\gamma = 53.4812 \pm 4.30599$
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $150 \leq E \leq 160$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

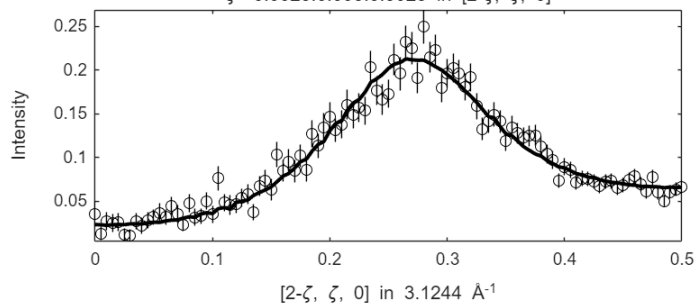


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $150 \leq E \leq 160$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

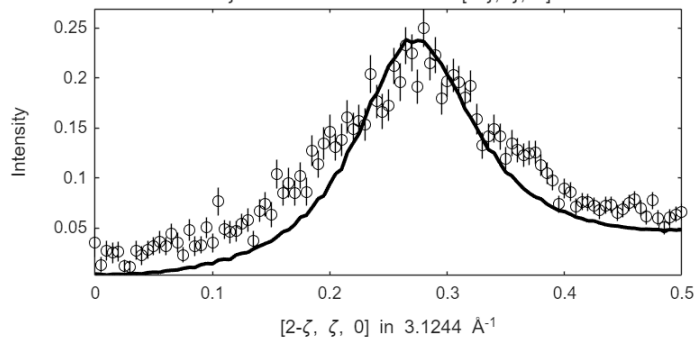


**** En = 165±5 going to 185

Fe_ei401_align_sym3.sqw
 $S = 1.2 \pm 0.065$; $J_0 = 36 \pm 0.44$; $J_1 = 0 \pm 0$; $\gamma = 56.8935 \pm 3.67395$
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $160 \leq E \leq 170$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



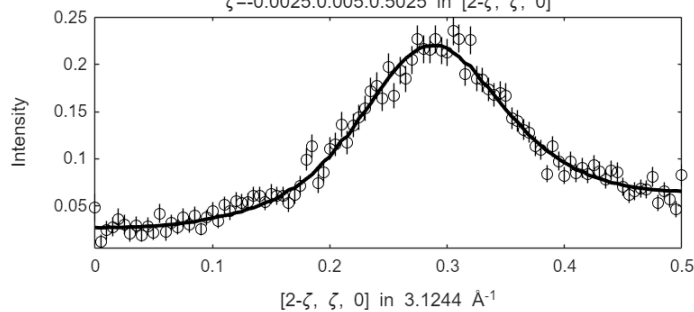
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $160 \leq E \leq 170$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



**** En = 175±5 going to 185

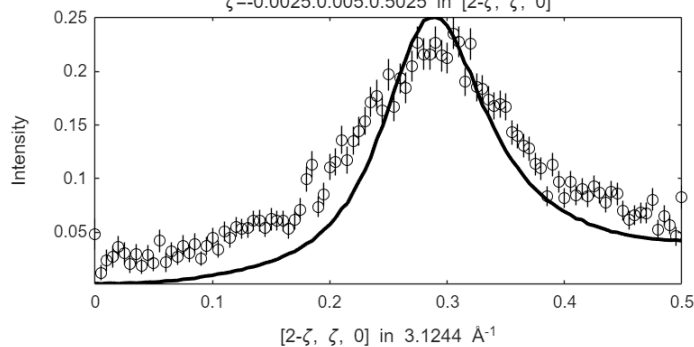
Fe_ei401_align_sym3.sqw

S= 1 ± 0.039 ; J0= 35 ± 0.23 ; J1= 0 ± 0 ; gamma= 44.962 ± 2.0632
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $170 \leq E \leq 180$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



Fe_ei401_align_sym3.sqw

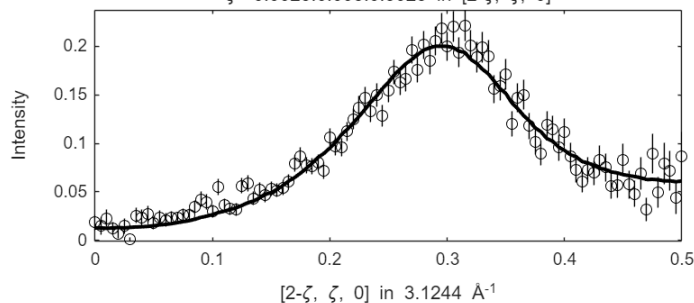
$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $170 \leq E \leq 180$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

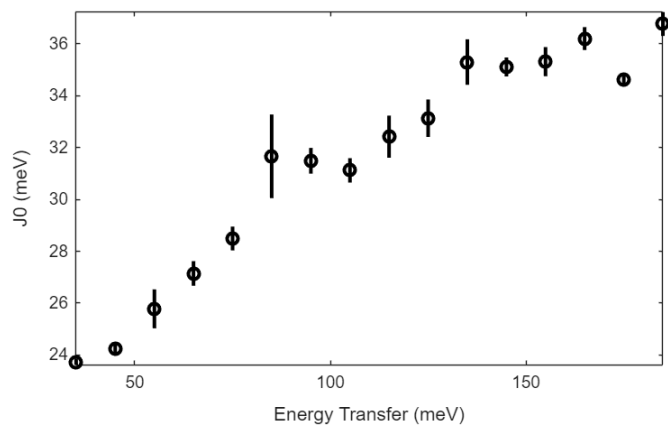
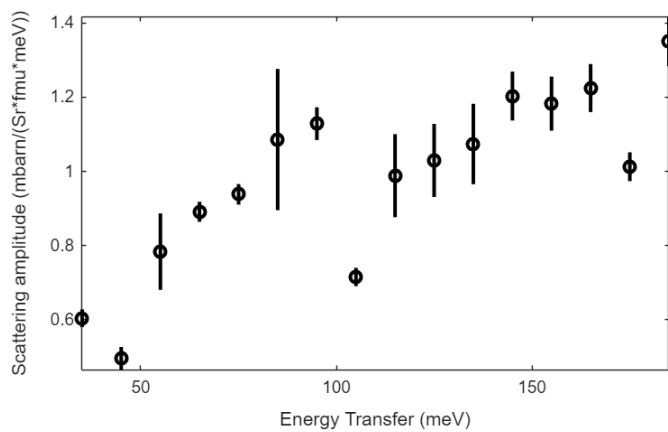
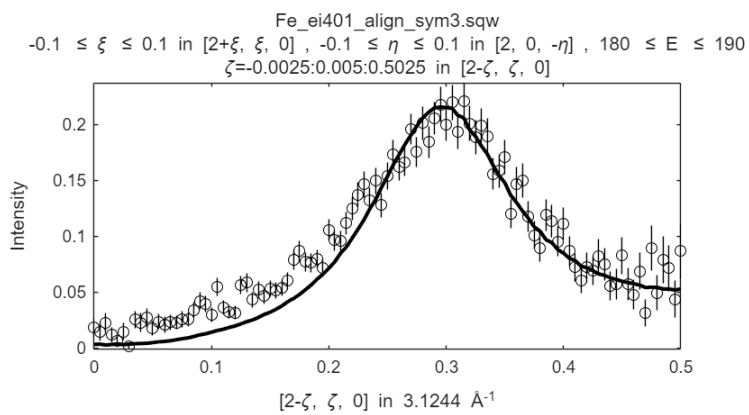


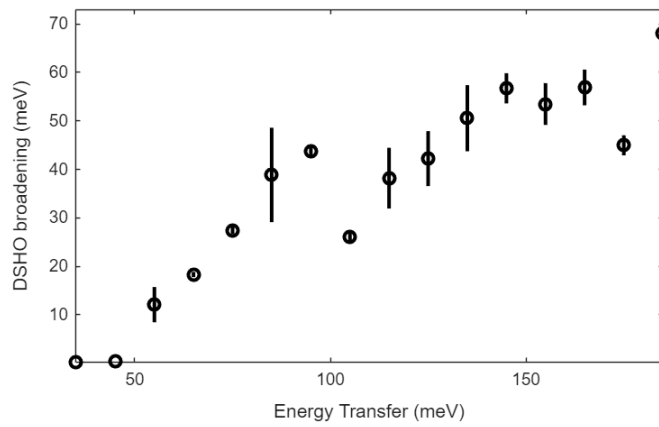
 **** En = 185 ± 5 going to 185

Fe_ei401_align_sym3.sqw

S= 1.3 ± 0.066 ; J0= 37 ± 0.46 ; J1= 0 ± 0 ; gamma= 68.1308 ± 4.74379
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $180 \leq E \leq 190$
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$







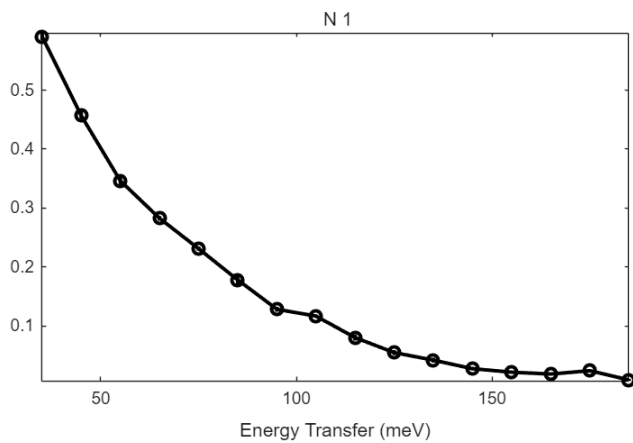
```

cl = NG_FFCutsFits.(selectCuts);
NG_FFCutsFits.(selectCuts) = [cl{:}];

beep
beep
beep

all_fit_par = NG_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

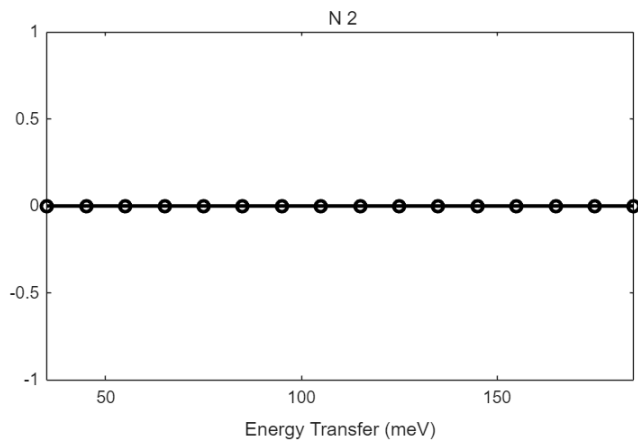
```



```

pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure

```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_NG.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_NG_',selectCuts,'.pdf']);
export(thisFile,targ_file);
%close all;
%end
```

```
fit_src_struc = NG_FFCutsFits;
save(Cutfitfile,"fit_src_struc","-v7.3')
```